Water electrolysis

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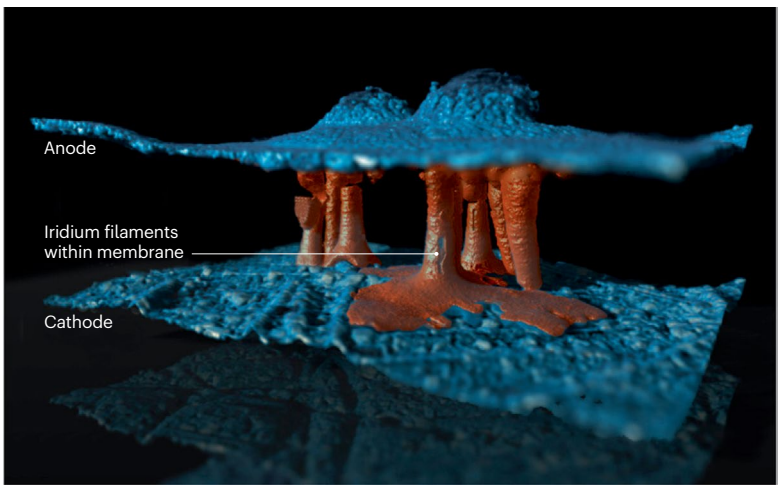
Degradation through filaments

Proton exchange membrane water electrolysers (PEMWEs) hold promise for green hydrogen fuel production from water, powered by renewable electricity. The successful deployment of PEMWEs requires large stacks that can operate for long stretches of time and under eventual intermittent conditions with minimal degradation. It is thus imperative to understand all potential degradation mechanisms under such relevant operational conditions. However, to date most studies focus on laboratory-scale setups under relatively mild conditions.

Now, in a joint effort between academia and industry, as part of the German H_2Giga initiative, Christine Heume and colleagues assess the degradation of a PEMWE stack under industrially relevant conditions during 5,000 h of operation, including 155 start-up/shut-down events. The researchers identify the cross-plane formation of metallic nanocrystalline iridium filaments growing across the membrane, which create parasitic current pathways that induce degradation and reduce the system's efficiency.



The location and morphology of the iridium filaments are characterized by X-ray computed tomography and focused ion beam scanning electron microscopy with energy-dispersive X-ray spectroscopy. Filaments form below weld seams on the porous transport layer at the anode and cross the membrane to the cathode. Other techniques such as Raman or electrochemical impedance spectroscopies are used to confirm their composition and properties. A possible formation mechanism is proposed by the researchers based on the available data, where weld seams would act as nucleation points for dissolved Ir^{3+} species, which are then reduced by H_2 crossover.

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